

ELITE IGCSE ACADEMY

Private Past Paper Solutions

May 2021 P2H Worked Solutions

May 2021 | P2H

27 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

May 2021 P2H - May 2021 | P2H

- 1** Write 600 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Prime factors, HCF and LCM. The tag is correct.**Method**

$$600 = 6 \times 100$$

$$600 = (2 \times 3)(2^2 \times 5^2)$$

$$600 = 2^3 \times 3 \times 5^2$$

FINAL ANSWER

$$2^3 \times 3 \times 5^2$$

PAST PAPER QUESTION**Q#: 2****Marks: 3**TOPIC: **Fractions**

May 2021 P2H - May 2021 | P2H

2 Show that $2\frac{4}{7} \div 1\frac{1}{8} = 2\frac{2}{7}$

(Total for Question 2 is 3 marks)



PAST PAPER SOLUTION**Q#: 2****Marks: 3**TOPIC: **Fractions**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Fractions. The tag is correct.**Method**

$$2\frac{4}{7} = \frac{18}{7}$$

$$1\frac{1}{8} = \frac{9}{8}$$

Divide by multiplying by the reciprocal:

$$\begin{aligned}\frac{18}{7} \div \frac{9}{8} &= \frac{18}{7} \times \frac{8}{9} \\ &= \frac{16}{7} = 2\frac{2}{7}\end{aligned}$$

FINAL ANSWER

$$2\frac{2}{7}$$

PAST PAPER QUESTION**Q#: 3****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

May 2021 P2H - May 2021 | P2H

- 3 The bearing of Paris from London is 149°
Work out the bearing of London from Paris.

.....
(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Bearings, Scale Drawing & Constructions. The tag is correct.**Solution**

The bearing of Paris from London is the reverse bearing of London from Paris.
Reverse bearings differ by 180° :

$$149^\circ + 180^\circ = 329^\circ$$

FINAL ANSWER **329° .**

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

May 2021 P2H - May 2021 | P2H

4 $\mathcal{E} = \{\text{letters of the alphabet}\}$ $B = \{\text{b, r, a, z, i, l}\}$ $I = \{\text{i, r, e, l, a, n, d}\}$

(a) List the members of the set

(i) $B \cup I$ (ii) $B \cap I'$

(2)

 $K = \{\text{k, e, n, y, a}\}$

Cody writes down the statement $B \cap K = \emptyset$
 Cody's statement is wrong.

(b) Explain why.

(1)

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Set notation and Venn diagrams. The tag is correct.**Method**

$$B = \{b, r, a, z, i, l\}$$

$$I = \{i, r, e, l, a, n, d\}$$

The union contains every letter that is in B or I :

$$B \cup I = \{a, b, d, e, i, l, n, r, z\}$$

$B \cap I'$ means letters in B , but not in I :

$$B \cap I' = \{b, z\}$$

For $K = \{k, e, n, y, a\}$, Cody is not correct because the letter a is in both B and K .

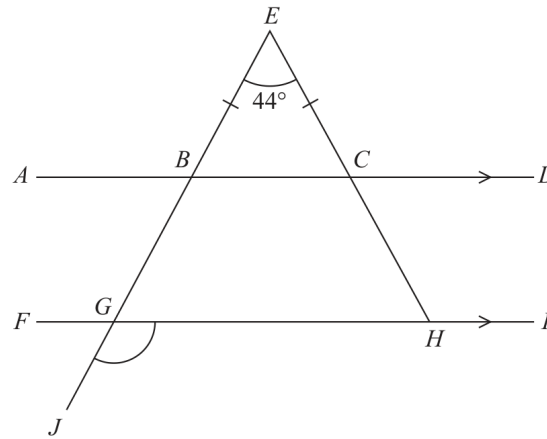
$$B \cap K = \{a\}$$

FINAL ANSWER

(a)(i) $\{a, b, d, e, i, l, n, r, z\}$, (a)(ii) $\{b, z\}$. Cody is incorrect.

PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

May 2021 P2H - May 2021 | P2H

5Diagram **NOT**
accurately drawn

$ABCD$ and $FGHI$ are parallel straight lines.
 $EBGJ$ and ECH are straight lines.

$BE = CE$

Angle $BEC = 44^\circ$

Work out the size of angle JGH .

Give a reason for each stage of your working.

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to angles in polygons and parallel lines.**Solution**Triangle EBC is isosceles because

$$BE = CE$$

So the base angles are equal.

$$\angle EBC = \angle ECB = \frac{180^\circ - 44^\circ}{2} = 68^\circ$$

The two horizontal lines are parallel, so the corresponding acute angle at G is 68° .

The required angle is supplementary to this angle:

$$180^\circ - 68^\circ = 112^\circ$$

FINAL ANSWER 112° .

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Percentages**

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6 Mariana sells bags of bird food.

The bags that Mariana sold last week each contained 12 kg of seeds.

The bags that she is going to sell next week will each contain a mixture of nuts and seeds where for each bag

$$\text{weight of nuts : weight of seeds} = 4 : 5$$

The total weight of the nuts and the seeds in each bag will be 19.35 kg

The weight of seeds in each bag that Mariana sells next week will be less than the weight of seeds in each bag that Mariana sold last week.

Work out this decrease as a percentage of the weight of seeds in each bag that Mariana sold last week.

Give your answer correct to one decimal place.

..... %

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 4**TOPIC: **Percentages**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct, with ratio used to find the new seed weight.**Method**

In each new bag, the ratio

$$\text{nuts : seeds} = 4 : 5$$

has 9 total parts.

Weight of seeds in each new bag:

$$\frac{5}{9} \times 19.35 = 10.75 \text{ kg}$$

Decrease in seed weight:

$$12 - 10.75 = 1.25 \text{ kg}$$

Percentage decrease:

$$\frac{1.25}{12} \times 100 = 10.416 \dots \%$$

FINAL ANSWER

10.4%

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

May 2021 P2H - May 2021 | P2H

7 Here is a right-angled triangle.

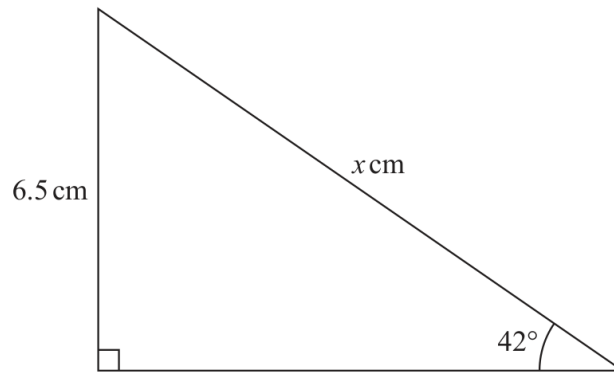


Diagram **NOT**
accurately drawn

Work out the value of x .

Give your answer correct to one decimal place.

$x =$

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.**Solution**

The side 6.5 cm is opposite the 42° angle and x is the hypotenuse.

$$\sin 42^\circ = \frac{6.5}{x}$$

$$x = \frac{6.5}{\sin 42^\circ} = 9.714 \dots$$

FINAL ANSWER $x = 9.7$ cm.

PAST PAPER QUESTION**Q#: 8****Marks: 3**TOPIC: **Simultaneous Equations**

May 2021 P2H - May 2021 | P2H

8 Solve the simultaneous equations

$$5a + 2c = 10$$

$$2a - 4c = 7$$

Show clear algebraic working.

$$a = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Simultaneous Equations**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Simultaneous equations. The tag is correct.**Solution**

$$5a + 2c = 10$$

$$2a - 4c = 7$$

Double the first equation:

$$10a + 4c = 20$$

Add this to the second equation:

$$12a = 27$$

$$a = \frac{9}{4}$$

Substitute into $5a + 2c = 10$:

$$5\left(\frac{9}{4}\right) + 2c = 10$$

$$\frac{45}{4} + 2c = \frac{40}{4}$$

$$2c = -\frac{5}{4}$$

$$c = -\frac{5}{8}$$

FINAL ANSWER

$$a = \frac{9}{4}, c = -\frac{5}{8}.$$

PAST PAPER QUESTION**Q#: 9****Marks: 3**TOPIC: **Factorising**

May 2021 P2H - May 2021 | P2H

9 (i) Factorise $x^2 + 2x - 24$

.....
(2)

(ii) Hence solve $x^2 + 2x - 24 = 0$

.....
(1)

(Total for Question 9 is 3 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Factorising**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Factorising. The tag is correct.**Solution**

$$x^2 + 2x - 24 = (x + 6)(x - 4)$$

So

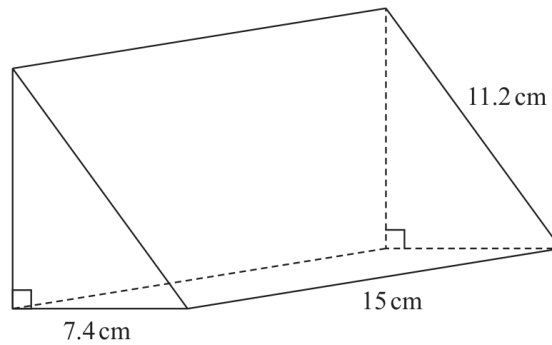
$$(x + 6)(x - 4) = 0$$

FINAL ANSWER

$$x = -6 \text{ or } x = 4$$

PAST PAPER QUESTIONTOPIC: **Volume & Surface Area**

May 2021 P2H - May 2021 | P2H

Q#: 10**Marks: 5****10** Here is a triangular prism.Diagram **NOT**
accurately drawn

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

..... cm³**(Total for Question 10 is 5 marks)**

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Volume & Surface Area**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Volume & Surface Area. The tag is correct.**Solution**

Find the height of the triangular cross-section:

$$h^2 = 11.2^2 - 7.4^2$$

$$h = 8.407 \dots$$

Area of the triangular cross-section:

$$\frac{1}{2}(7.4)(8.407 \dots) = 31.106 \dots$$

Volume of the prism:

$$31.106 \dots \times 15 = 466.596 \dots$$

FINAL ANSWER467 cm³.

PAST PAPER QUESTION**Q#: 11****Marks: 6****TOPIC: Compound Interest & Depreciation**

May 2021 P2H - May 2021 | P2H

11 Chengbo sold a house for 180 000 yuan.

The amount for which he sold the house is 24% more than the amount he paid for the house.

- (a) Work out how much Chengbo paid for the house.
Give your answer correct to 3 significant figures.

..... yuan
(3)

Zhi bought a house on 1st January 2017

When she bought the house, its value was 120 000 yuan.

The value of the house increased by 1.8% per year.

- (b) Work out the value of Zhi's house on 1st January 2020
Give your answer correct to 3 significant figures.

..... yuan
(3)

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 6****TOPIC: Compound Interest & Depreciation**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to Compound Interest and Depreciation. The repeated yearly percentage increase is the key skill.

Method

For Chengbo's house, 180000 yuan is 124% of the price paid.

$$\text{price paid} = 180000 \div 1.24 = 145161.29 \dots$$

To 3 significant figures:

$$145000$$

For Zhi's house, the value increases by 1.8% each year for 3 years.

$$120000 \times 1.018^3 = 126598.2 \dots$$

To 3 significant figures:

$$127000$$

FINAL ANSWER

Chengbo paid 145000 yuan. Zhi's house was worth 127000 yuan.

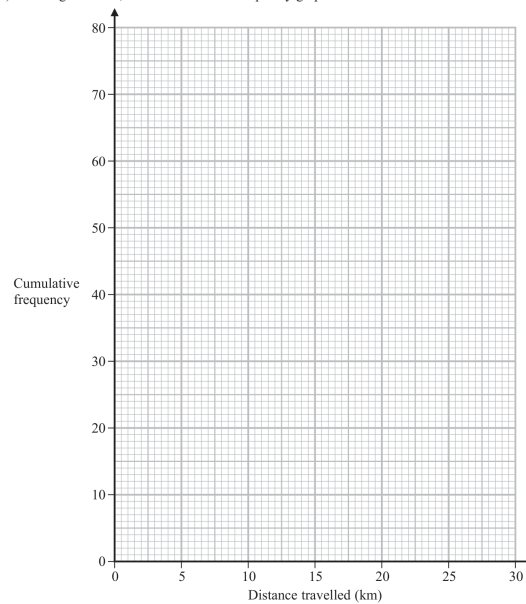
PAST PAPER QUESTION**Q#: 12****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

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- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

1

.....

.....

2

.....

.....

(2)

(Total for Question 12 is 7 marks)

PAST PAPER SOLUTION**Q#: 12** **Marks: 7****TOPIC: Cumulative Frequency Diagrams**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Cumulative Frequency Diagrams. The tag is correct.**Solution**

For Office A, the cumulative frequencies are

$$17, 32, 57, 70, 76, 80$$

The median is the 40th value:

$$10 + \frac{40 - 32}{57 - 32} \times 5 = 11.6$$

The quartiles are the 20th and 60th values:

$$Q_1 = 5 + \frac{20 - 17}{32 - 17} \times 5 = 6.0$$

$$Q_3 = 15 + \frac{60 - 57}{70 - 57} \times 5 \approx 16.2$$

$$\text{IQR} \approx 16.2 - 6.0 = 10.2$$

Office A has a smaller median than Office B, so the typical journey is shorter.

Office A has a larger IQR, so the distances are more spread out.

FINAL ANSWER

median 11.6 km, IQR about 10.2 km.

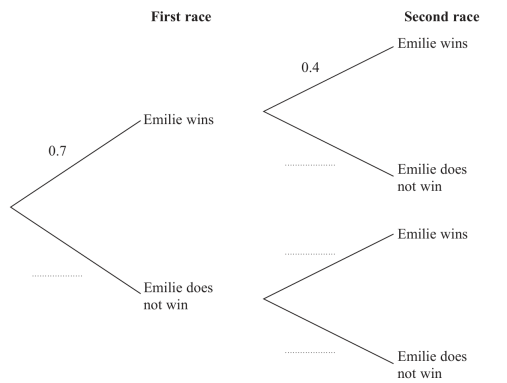
PAST PAPER QUESTION**Q#: 13****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

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13 Emilie takes part in two races.

The probability that she wins the first race is 0.7
 The probability that she wins the second race is 0.4
 The outcomes of the two races are independent.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Emilie wins exactly one of the two races.

(3)

Emilie is going to take part in a third race.

If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

(3)

(Total for Question 13 is 8 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

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WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Diagrams - Venn & Tree Diagrams. The tag is correct.**Solution**

For the two races:

$$P(W_1) = 0.7, \quad P(L_1) = 0.3$$

$$P(W_2) = 0.4, \quad P(L_2) = 0.6$$

Exactly one win:

$$0.7 \times 0.6 + 0.3 \times 0.4$$

$$= 0.42 + 0.12 = 0.54$$

For exactly two wins in three races:

First two wins, then lose the third:

$$0.7 \times 0.4 \times 0.4 = 0.112$$

Exactly one win in the first two, then win the third:

$$0.54 \times 0.3 = 0.162$$

$$0.112 + 0.162 = 0.274$$

FINAL ANSWER

0.54, 0.274.

PAST PAPER QUESTION**Q#: 14****Marks: 3**TOPIC: **Algebraic Roots & Indices**

May 2021 P2H - May 2021 | P2H

14 Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^{-2}$

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 3****TOPIC: Algebraic Roots & Indices**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic Roots and Indices. The tag is correct.**Method**

$$\left(\frac{9x^4}{16y^{10}} \right)^{-1/2}$$

A negative power takes the reciprocal:

$$\left(\frac{16y^{10}}{9x^4} \right)^{1/2}$$

Now take the square root of each factor:

$$\frac{\sqrt{16y^{10}}}{\sqrt{9x^4}} = \frac{4y^5}{3x^2}$$

FINAL ANSWER

$$\frac{4y^5}{3x^2}.$$

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Graphs of Functions**

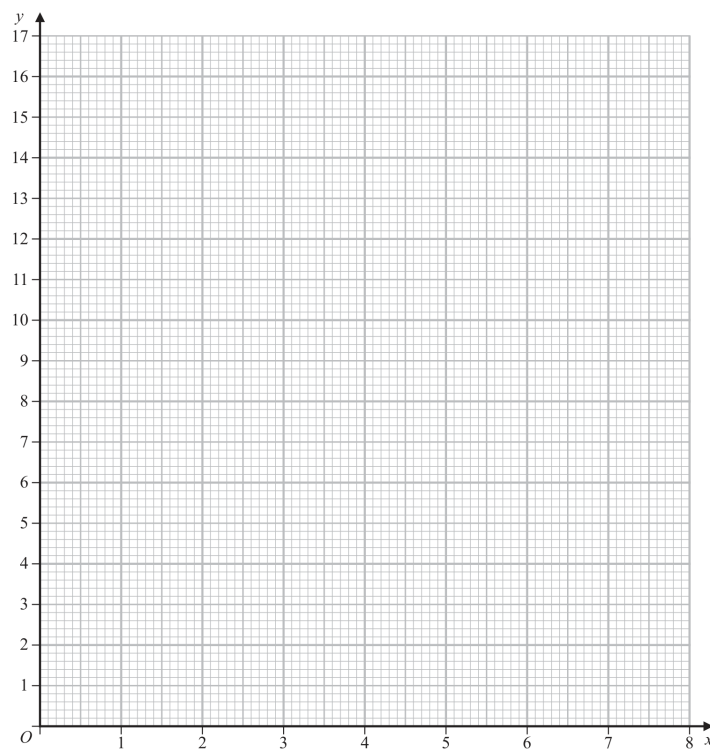
May 2021 P2H - May 2021 | P2H

- 15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

- (b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leq x \leq 8$



(2)

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 4****TOPIC: Graphs of Functions**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This is a table-and-graph question for $x + \frac{4}{x}$.**Solution**

$$y = \frac{1}{x}(x^2 + 4) = x + \frac{4}{x}$$

The completed table is:

x	0.25	0.5	1	2	4	8
y	16.25	8.5	5	4	5	8.5

Plot the points and join them with a smooth curve.

FINAL ANSWER

Missing values are 8.5, 5, 4, 5.

PAST PAPER QUESTION**Q#: 16****Marks: 6****TOPIC: Direct & Inverse Proportion**

May 2021 P2H - May 2021 | P2H

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r
(3)(b) Find the value of A when $r = 7.5A$

$$A = \text{.....}$$

(3)

(Total for Question 16 is 6 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 6****TOPIC: Direct & Inverse Proportion**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Direct and Inverse Proportion. The tag is correct.**Method** A is inversely proportional to r^2 , so

$$A = \frac{k}{r^2}$$

Use $A = 5$ when $r = 0.3$:

$$5 = \frac{k}{0.3^2}$$

$$k = 5 \times 0.09 = 0.45$$

So

$$A = \frac{0.45}{r^2}$$

For part (b), $r = 7.5A$. Substitute into the formula:

$$A = \frac{0.45}{(7.5A)^2}$$

$$A = \frac{0.45}{56.25A^2}$$

$$56.25A^3 = 0.45$$

$$A^3 = 0.008$$

$$A = 0.2$$

FINAL ANSWER(a) $A = \frac{0.45}{r^2}$, (b) $A = 0.2$.

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Coordinate Geometry**

May 2021 P2H - May 2021 | P2H

17 The straight line **L** passes through the points $(4, -1)$ and $(6, 4)$

The straight line **M** is perpendicular to **L** and intersects the y -axis at the point $(0, 8)$

Find the coordinates of the point where **M** intersects the x -axis.

(.....,)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 4**TOPIC: **Coordinate Geometry**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Coordinate geometry. The tag is correct.**Solution**Line L passes through $(4, -1)$ and $(6, 4)$.

Its gradient is

$$\frac{4 - (-1)}{6 - 4} = \frac{5}{2}$$

Line M is perpendicular to L , so its gradient is

$$-\frac{2}{5}$$

Line M intersects the y-axis at $(0, 8)$, so

$$y = -\frac{2}{5}x + 8$$

At the x-axis, $y = 0$:

$$0 = -\frac{2}{5}x + 8$$

$$\frac{2}{5}x = 8$$

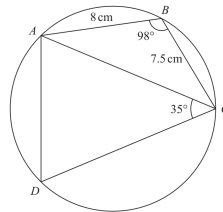
$$x = 20$$

FINAL ANSWER $(20, 0)$.

PAST PAPER QUESTION**Q#: 18****Marks: 6****TOPIC: Circle Theorems**

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18

Diagram **NOT**
accurately drawn $ABCD$ is a quadrilateral where A , B , C and D are points on a circle. $AB = 8$ cm $BC = 7.5$ cmAngle $ABC = 98^\circ$ Angle $ACD = 35^\circ$ Work out the perimeter of quadrilateral $ABCD$.
Give your answer correct to one decimal place.

_____ cm

(Total for Question 18 is 6 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 6**TOPIC: **Circle Theorems**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Circle Theorems. The cyclic quadrilateral theorem is needed.**Solution**In triangle ABC , use the cosine rule:

$$AC^2 = 8^2 + 7.5^2 - 2(8)(7.5) \cos 98^\circ$$

$$AC = 11.702 \dots$$

Opposite angles in a cyclic quadrilateral add to 180° , so

$$\angle ADC = 180^\circ - 98^\circ = 82^\circ$$

In triangle ACD ,

$$\angle CAD = 180^\circ - 82^\circ - 35^\circ = 63^\circ$$

Use the sine rule:

$$AD = \frac{AC \sin 35^\circ}{\sin 82^\circ} = 6.778 \dots$$

$$CD = \frac{AC \sin 63^\circ}{\sin 82^\circ} = 10.529 \dots$$

Perimeter:

$$8 + 7.5 + 6.778 \dots + 10.529 \dots = 32.807 \dots$$

FINAL ANSWER

32.8 cm.

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

May 2021 P2H - May 2021 | P2H

19 Solve the simultaneous equations

$$\begin{aligned}y &= 3 - 2x \\ x^2 + y^2 &= 18\end{aligned}$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Simultaneous equations. The tag is correct.**Solution**

$$y = 3 - 2x$$

$$x^2 + y^2 = 18$$

Substitute $y = 3 - 2x$:

$$x^2 + (3 - 2x)^2 = 18$$

$$x^2 + 9 - 12x + 4x^2 = 18$$

$$5x^2 - 12x - 9 = 0$$

Factorise:

$$(5x + 3)(x - 3) = 0$$

So

$$x = -\frac{3}{5} \quad \text{or} \quad x = 3$$

If $x = -\frac{3}{5}$, then

$$y = 3 - 2\left(-\frac{3}{5}\right) = \frac{21}{5}$$

If $x = 3$, then

$$y = 3 - 6 = -3$$

FINAL ANSWER

$$(x, y) = \left(-\frac{3}{5}, \frac{21}{5}\right) \text{ or } (3, -3).$$

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May 2021 P2H - May 2021 | P2H

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is 300 cm^3

Given that

the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The surface area ratio of the small block to the large block is

$$25 : 36$$

So the linear ratio is

$$\sqrt{25} : \sqrt{36} = 5 : 6$$

Therefore the volume ratio is

$$5^3 : 6^3 = 125 : 216$$

The small block has volume 300 cm^3 .

Large block volume:

$$300 \times \frac{216}{125} = 518.4$$

FINAL ANSWER

$$518.4 \text{ cm}^3$$

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May 2021 P2H - May 2021 | P2H

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is 300 cm^3

Given that

the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The surface area ratio of the small block to the large block is

$$25 : 36$$

So the linear ratio is

$$\sqrt{25} : \sqrt{36} = 5 : 6$$

Therefore the volume ratio is

$$5^3 : 6^3 = 125 : 216$$

The small block has volume 300 cm^3 .

Large block volume:

$$300 \times \frac{216}{125} = 518.4$$

FINAL ANSWER

$$518.4 \text{ cm}^3$$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Differentiation**

May 2021 P2H - May 2021 | P2H

- 21** The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Differentiation**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$y = kx^2 + \frac{16}{x}$$

Differentiate:

$$\frac{dy}{dx} = 2kx - \frac{16}{x^2}$$

At the stationary point, $x = \frac{2}{3}$, so the gradient is 0:

$$2k \left(\frac{2}{3} \right) - \frac{16}{\left(\frac{2}{3} \right)^2} = 0$$

$$\frac{4k}{3} - \frac{16}{\frac{4}{9}} = 0$$

$$\frac{4k}{3} - 36 = 0$$

$$\frac{4k}{3} = 36$$

$$k = 27$$

Now find a , the y -coordinate when $x = \frac{2}{3}$:

$$a = 27 \left(\frac{2}{3} \right)^2 + \frac{16}{\frac{2}{3}}$$

$$a = 27 \cdot \frac{4}{9} + 24$$

$$a = 12 + 24 = 36$$

FINAL ANSWER

$$a = 36$$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Differentiation**

May 2021 P2H - May 2021 | P2H

- 21** The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Differentiation**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$y = kx^2 + \frac{16}{x}$$

Differentiate:

$$\frac{dy}{dx} = 2kx - \frac{16}{x^2}$$

At the stationary point, $x = \frac{2}{3}$, so the gradient is 0:

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$$a = 27 \left(\frac{2}{3} \right)^2 + \frac{16}{\frac{2}{3}}$$

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$$a = 12 + 24 = 36$$

FINAL ANSWER

$$a = 36$$

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Completing the Square**

May 2021 P2H - May 2021 | P2H

22 The curve **S** has equation $y = f(x)$ where $f(x) = x^2$ The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$ By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants, describe fully a series of transformations that map the curve **S** onto the curve **T**.

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(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Completing the Square**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$g(x) = 2x^2 - 12x + 13$$

$$= 2(x^2 - 6x) + 13$$

$$= 2((x - 3)^2 - 9) + 13$$

$$= 2(x - 3)^2 - 18 + 13$$

$$= 2(x - 3)^2 - 5$$

Starting with $S : y = x^2$:

- Translate 3 units to the right to get $y = (x - 3)^2$.
- Stretch parallel to the y -axis by scale factor 2 to get $y = 2(x - 3)^2$.
- Translate 5 units down to get $y = 2(x - 3)^2 - 5$.

FINAL ANSWER

$g(x) = 2(x - 3)^2 - 5$. Transformations: right 3, stretch parallel to the y -axis by scale factor 2, then down 5.

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Completing the Square**

May 2021 P2H - May 2021 | P2H

- 22** The curve **S** has equation $y = f(x)$ where $f(x) = x^2$
 The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants,
 describe fully a series of transformations that map the curve **S** onto the curve **T**.

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(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Completing the Square**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

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Starting with $S : y = x^2$:

- Translate 3 units to the right to get $y = (x - 3)^2$.
- Stretch parallel to the y -axis by scale factor 2 to get $y = 2(x - 3)^2$.
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FINAL ANSWER

$g(x) = 2(x - 3)^2 - 5$. Transformations: right 3, stretch parallel to the y -axis by scale factor 2, then down 5.

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P2H - May 2021 | P2H

23 Pippa has a box containing N pens.

There are only black pens and red pens in the box.

The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of N .

Show clear algebraic working.

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**Let the number of red pens be r .

Then the number of black pens is

$$r + 3$$

and

$$N = 2r + 3$$

So

$$r = \frac{N - 3}{2}$$

and

$$r + 3 = \frac{N + 3}{2}$$

The probability of black followed by red is

$$\frac{(N + 3)/2}{N} \cdot \frac{(N - 3)/2}{N - 1} = \frac{9}{35}$$

$$\frac{N^2 - 9}{4N(N - 1)} = \frac{9}{35}$$

$$35(N^2 - 9) = 36N(N - 1)$$

$$35N^2 - 315 = 36N^2 - 36N$$

$$N^2 - 36N + 315 = 0$$

$$(N - 15)(N - 21) = 0$$

FINAL ANSWER

$$N = 15 \text{ or } N = 21$$

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P2H - May 2021 | P2H

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There are only black pens and red pens in the box.

The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$ Find the possible values of N .

Show clear algebraic working.

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P2H - May 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**Let the number of red pens be r .

Then the number of black pens is

$$r + 3$$

and

$$N = 2r + 3$$

So

$$r = \frac{N - 3}{2}$$

and

$$r + 3 = \frac{N + 3}{2}$$

The probability of black followed by red is

$$\frac{(N + 3)/2}{N} \cdot \frac{(N - 3)/2}{N - 1} = \frac{9}{35}$$

$$\frac{N^2 - 9}{4N(N - 1)} = \frac{9}{35}$$

$$35(N^2 - 9) = 36N(N - 1)$$

$$35N^2 - 315 = 36N^2 - 36N$$

$$N^2 - 36N + 315 = 0$$

$$(N - 15)(N - 21) = 0$$

FINAL ANSWER

$$N = 15 \text{ or } N = 21$$